

POWER

File: power.kicad_sch

ESP32-C6

File: esp32-c6.kicad_sch

EXPANSION

File: expansion.kicad_sch

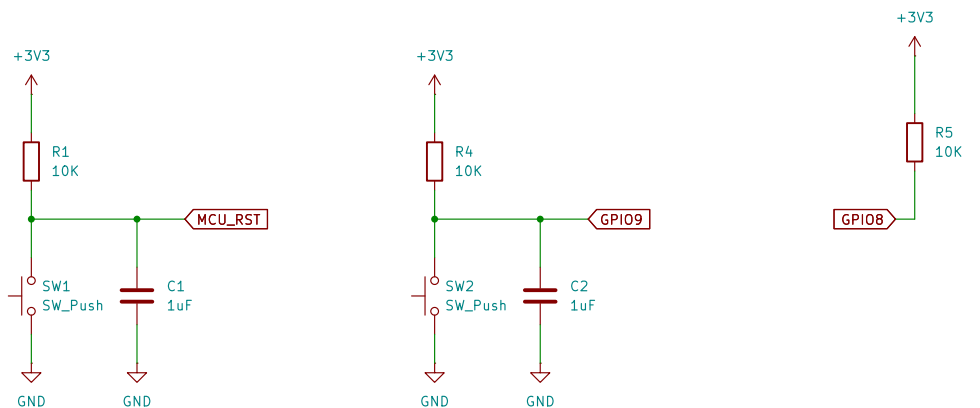
Approved By: Jeff Mboya
Approved By: Jones Kisaka
Designed By: Osodo Rodney
Abstract Machines

Sheet: /
File: a0.kicad_sch

Title: A0

Size: A4	Date: 2026-07-09	Rev: v0.5.0
KiCad E.D.A. 9.0.2		Id: 1/4

PULL UP

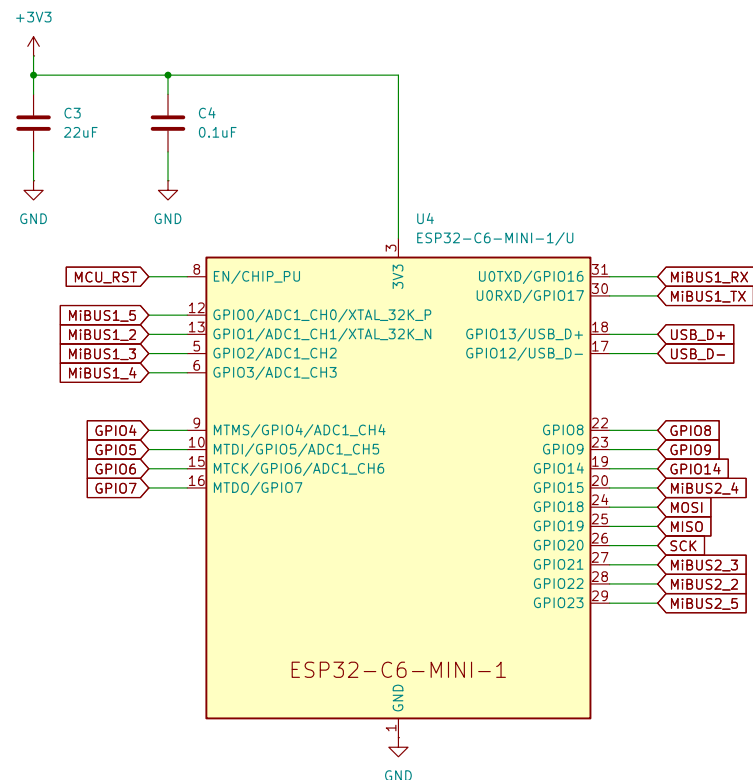


EN – MCU_RST
High: on, enables the chip.
Low: off, the chip powers off.

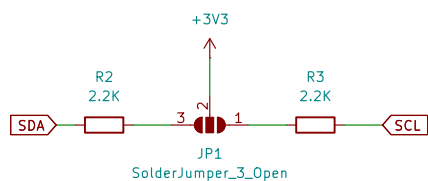
Strapping pin: GPIO8 and GPIO9

SPI boot: GPIO8 – Any value GPIO9 – 1
Joint download boot: GPIO8 – 1 GPIO9 – 0

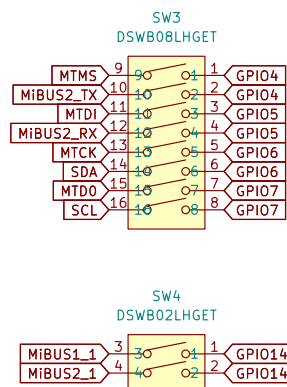
ESP32C6



I2C PULLUPS



DIP SWITCH



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Sheet: /ESP32-C6/
File: esp32-c6.kicad_sch

Title: A0

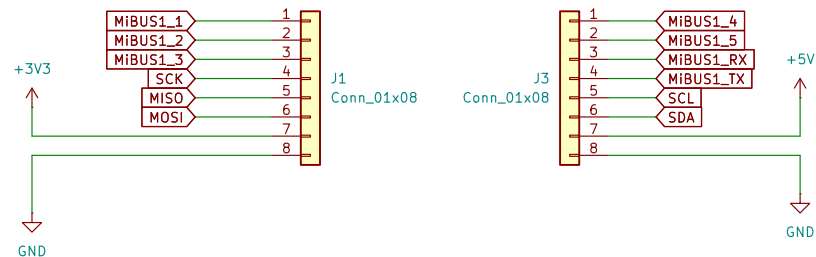
Size: A4 Date: 2026-07-09

KiCad E.D.A. 9.0.2

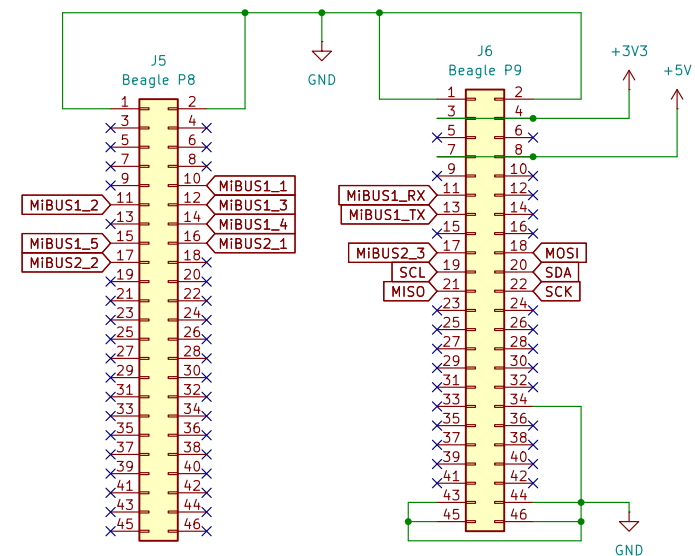
Rev: v0.5.0

Id: 2/4

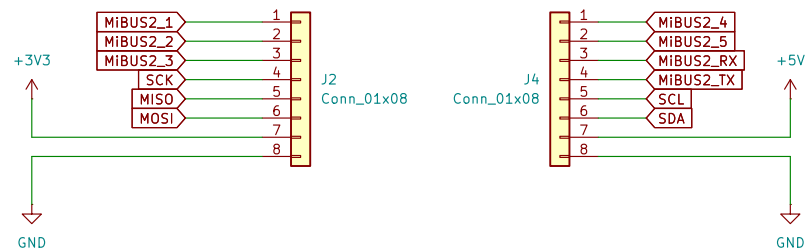
MiBUS1 CONNECTOR



HEADER PINS



MiBUS2 CONNECTOR



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Abstract Machines

Sheet: /EXPANSION/

File: expansion.kicad_sch

Title: A0

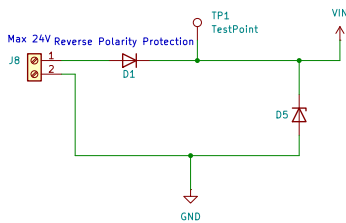
Size: A4 Date: 2026-07-09

KiCad E.D.A. 9.0.2

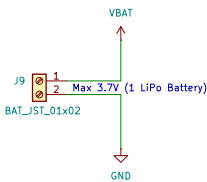
Rev: v0.5.0

Id: 4/4

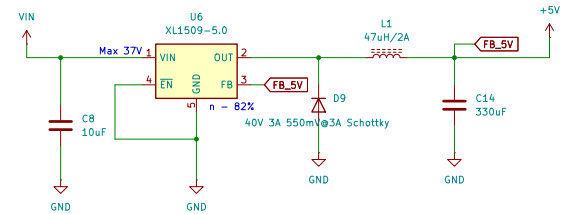
REVERSE POLARITY PROTECTION



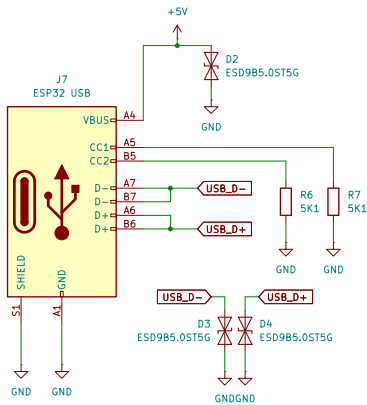
BATTERY



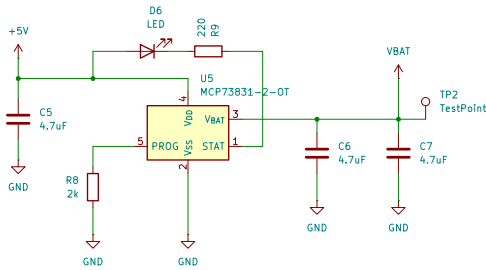
VIN – 5V CONV



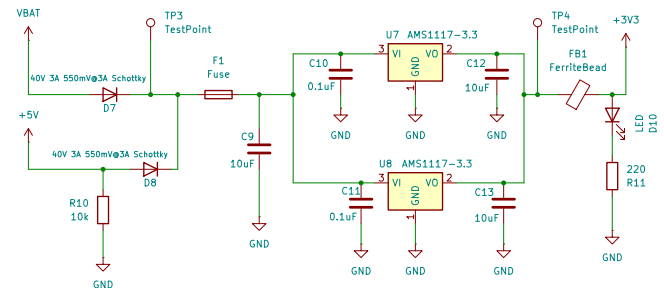
USB C & PWR IN



BATTERY CHARGER



5V – 3.3V CONV



CURRENT REGULATION SET
Fast charge current regulation can be scaled by placing a programming resistor (RPROG) from the PROG input to VSS. The program resistor and the charge current are calculated using the following equation:
 $I_{reg} = 1000V/R_{prog}$
where: Rprog is in kOhms
Ireg is charging current in milliamperes.
 $I_{reg} = 1000V/2 = 500mA$

